

James Wade

NSF Graduate Research Fellow

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EDUCATION

Bachelor of Mechanical Engineering; Minors in Computer Science & Mathematics 2019–2020, 2022–2026
Brigham Young University – Ira A. Fulton College of Engineering

- **GPA: 4.0/4.0** (Rank #1 of 485 in class) | Emphasis in controls & robotic systems | **GRE:** Quant 170/170, Verbal 166/170
- **Russell M. Nelson Scholar:** most prestigious scholarship at BYU, awarded to <1% of students (four-year, full-ride)
- **Groups:** Mars Rover Team Lead, Rocketry Club, Spacecraft Club, Mechatronics Club, Global Engineering Outreach
- *Note: took a hiatus from studies 2020–2022 to serve a voluntary service mission in Mongolia*

RESEARCH, INTERNSHIPS & TECHNICAL LEADERSHIP

Autonomy Intern May 2025 – Present
Sandia National Labs, AutonomyNM Division – Albuquerque, NM

- Designed **RAM-Stack**, a ROS-based autonomous multi-agent software stack for SLAM and navigation, integrating SLAM Toolbox, SwarmSLAM, Navigation2, and ROS 2 Control to enable coordinated mapping and path planning
- Diagnosed and optimized Navigation2 logic using **Behavior Tree** control flow (reactive sequences, fallbacks, decorators)
- Developing **VLA-based** behavioral policies for variable-formation drone swarms across simulation and hardware platforms (*ongoing*)

Safe Control Research Assistant Aug 2024 – Present
Brigham Young University, MAGICC Lab – Provo, UT

- Developing a discrete-time **Control Barrier Function** framework for bipedal robots using LiDAR projections in dynamic environments
- Using **conformal prediction** to provide probabilistic safety guarantees on the boundary of the safe region

Robotics Research Assistant – Team Lead, Founding Member Aug 2023 – Present
Brigham Young University, CMR Lab – Provo, UT
One of six national finalists, NASA BIG Idea Challenge

- Founding member of the lab; established technical infrastructure and led an undergraduate team to develop a lunar-deployable soft robot
- Architected the redesign of a modular inflatable robot, implementing a custom multi-node radio network with error-checking protocols (*manuscript in preparation for resubmission to **IEEE Transactions on Mechatronics***)
- Developing kinematic reconfiguration algorithms to maximize workspace and ensure fault tolerance under actuator failures (*manuscript submitted to **IEEE RA-L***)
- Secured follow-on funding from the Utah NASA Space Grant Consortium through submitted proposal

BYU Mars Rover Team Lead Aug 2025 – May 2026
Brigham Young University – Provo, UT

- Elected by peers to lead a **34-person team** competing in the 2025–2026 University Rover Challenge; manage technical subsystem leads across mechanical, electrical, and software divisions
- Oversaw system-level integration testing and served as primary technical resource for cross-subsystem debugging
- Architected **migration from ROS 1 to ROS 2**; developed training curriculum and led technical workshops to upskill team members in modern robotics software architecture

Embedded Systems Intern May 2024 – May 2025
Lawrence Livermore National Laboratory (LLNL) – Livermore, CA

- Engineered a custom, full-stack handheld controller (mechanical housing, circuit design, firmware) to teleoperate a fleet of autonomous marine rafts, replacing a cumbersome laptop-based system
- Programmed **FreeRTOS** task management and developed a custom LVGL-based control UI
- Delivered end-to-end mechanical, electrical, and software systems as the **sole developer** across the project lifecycle

Compliant Mechanisms Research Assistant

June 2023 – Aug 2023

Mark Rober / Brigham Young University, CMR Lab – Provo, UT

- Contributed to development of miniaturized compliant mechanisms for [The World's Smallest Nerf Gun](#) project (75M+ views on YouTube) across five size scales
- Conducted **100+ design iterations** to identify failure modes and optimize mechanical performance
- Collaborated in weekly technical meetings to drive design refinements based on experimental data

Microfluidics Research Assistant

March 2023 – Aug 2023

Brigham Young University, TEMP Lab – Provo, UT

- Co-authored paper in *International Journal of Heat and Mass Transfer* on lab-on-a-chip thermal sensing using quantum dot fluorescence
- Conducted fluorescence characterization experiments and analyzed thermal response data to validate temperature measurement accuracy across operating conditions

TEACHING EXPERIENCES

Robotics Teaching Assistant

Aug 2025 – Dec 2025

Brigham Young University – Provo, UT

- Selected as a TA for a **60-student** graduate-level course on manipulators, kinematics, and dynamics
- Mentored students through theoretical derivations and ROS/Python implementations during weekly office hours

Service Volunteer & English Teacher

June 2020 – June 2022

The Church of Jesus Christ of Latter-day Saints – Ulaanbaatar, Mongolia

- Completed a two-year full-time service mission, teaching English **12+ hours/week** in immigration offices, hospitals, and community centers
- Achieved conversational fluency in Mongolian

PUBLICATIONS & PRESENTATIONS

Publications

1. **James Wade**, Isaac Weaver, and Nathan Usevitch, “Fault-Tolerant, Rigidity-Preserving Control of Inflatable Truss Robots,” *IEEE Robotics and Automation Letters* [Submitted May 2026]
2. Mihai Stanciu, Isaac Weaver, Adam Rose, **James Wade**, Kaden Paxton, Chris Paul, Spencer Stowell, and Nathan Usevitch, “Modular Isoperimetric Soft Robotic Truss for Lunar Applications,” *IEEE Transactions on Mechatronics* [Submitted, February 2026; preparing for resubmission]
3. Derek Sanchez, Robert Macdonald, Brendan Mitchell, **James Wade**, McKay Wilkerson, Hunter Hinnen, Marshall Rawlins, Gregory P. Nordin, Adam T. Woolley, Troy R. Munro, “Advancing the applications of 3D printed microfluidics: Utilizing quantum dots to measure internal temperature,” *International Journal of Heat and Mass Transfer*, 2025

Conference Proceedings & Presentations — Podium

1. **James Wade**, “Fault-Tolerant, Rigidity-Preserving Control of Inflatable Truss Robots,” *Utah NASA Space Grant Consortium Symposium*, May 2025
2. **James Wade**, Chris Paul, Mihai Stanciu, Spencer Stowell, Isaac Weaver, Adam Rose, Ashleigh Cerven, Annie O'Bryan, Brian Bodily, Logan Yang, and Nathan Usevitch, “Untethered and Modular Inflatable Robot for Lunar Applications,” Finalists in the 2024 NASA BIG Idea Challenge — Inflatable Systems for Lunar Operations. Technical paper presented at the *NASA BIG Idea Symposium*, November 2024
3. **James Wade**, Chris Paul, “Constant-Pressure Untethered Soft Robotics: An Adaptable Solution to the Limitations of Soft Robots,” *Utah Conference on Undergraduate Research*, 2024 (announcement of NASA grant)

Conference Proceedings & Presentations — Poster

1. **James Wade**, “ROS 2 Autonomy Stack for Multi-Agent SLAM and Navigation,” presented at *Sandia New Mexico SIP Intern Symposium*, August 2025
2. **James Wade**, Chris Paul, “Untethered and Modular Inflatable Robot for Lunar Applications,” *Lunar Surface Innovation Consortium (LSIC)*, November 2024

SCHOLARSHIPS & GRANTS

2026 NSF Graduate Research Fellowship, \$137,000 in research funding over three years

- 2025 **Utah NASA Space Grant Consortium Fellowship**, \$2,000 research stipend for work on multiply-redundant soft robots in space environments; received one of **17** available awards at BYU (14 graduates, 3 undergraduates)
- 2024 **NASA BIG Idea Competition Funding**, secured \$140,000 as head grant writer and team lead; one of **six national finalists** in the 2024–2025 intercollegiate competition
- 2019 **President Russell M. Nelson Scholar**, four-year full-ride scholarship for 150% of tuition, awarded to fewer than **1%** of the student body (the highest scholarship available at BYU)

HONORS & AWARDS

- 2026 Awarded the **NSF Graduate Research Fellowship**
- 2025 Ranked **1st** in Mechanical Engineering class of 485
- 2025 Dean's List (all eight completed semesters), top **5%** of BYU Engineering department
- 2023 Phi Kappa Phi (honor society initiation), top **7.5%** of BYU Engineering juniors

SELECT TECHNICAL PROJECTS

Diffusion Models – From MNIST Generation to Robotic Control ([link](#))

- Reimplemented Diffusion Policy (Chi et al.) and Diffuser (Janner et al.) from scratch, building a four-stage progression from DDPM/DDIM image generation to action-diffusion robotic control and classifier-guided trajectory planning

Trebuchet Numerical Simulator – Graduate Dynamics Class Project ([link](#))

- Derived unified equations of motion using Euler-Lagrange formulation with Lagrangian multipliers to compute constraint forces across three contact modes: ground sliding, pendulum swing, and ballistic flight
- Implemented a hybrid dynamics simulator with event-triggered mode transitions and constraint relaxation to extract normal/friction forces from variational mechanics

RELEVANT COURSEWORK & SKILLS

Robotics & Control:	ROS 2 (Nav2, SLAM Toolbox, ROS 2 Control, MoveIt 2), ROS, Control Theory (PID, State-Space, Observers, MPC, LQR), State Estimation, Mechatronics, Dynamic System Modeling, Kinematics, Deep Learning
Mechanical Design & Analysis:	SolidWorks (Certified Professional, Sheet Metal), Ansys FEA, Mechanical System Design, Thermodynamics, GD&T, 3D Printing
Electrical & Embedded Systems:	Arduino, ESP32, STM32, FreeRTOS, Verilog, PCB Design
Programming:	C/C++, Python, PyTorch, Rust, MATLAB, Julia, Qt, Git, LaTeX, Java